

Design science build-evaluate cycle

ENVIRONMENT

People
analysts, marketers

Organisations
online retail

Technology
clickstream logs

Business need

Which behavioural
drivers matter,
at which point
in the journey,
and can the
answer be trusted?

DESIGN SCIENCE

BUILD

- Funnel-stage SHAP artefact
- stage cut-points + prefixes
- stage models per S_k
- three explanation layers

refine

EVALUATE

- PR-AUC lift, ROC-AUC, ECE
- faithfulness, stability
- seed + cross-paradigm
- bootstrap CIs, Holm

23 logged amendments
(3 cut-point redefinitions, 2 retractions)

KNOWLEDGE BASE

Foundations
Shapley values
TreeSHAP, TimeSHAP
journey theory

Methodologies
pre-registration
bootstrap CIs
DeLong, Holm
faithfulness metrics

relevance

rigour

application

additions

Contribution: stage conditioning is a modelling commitment, not a reporting convenience —
and explanation quality must be validated per stratum, not per model